DCE-KG: A Disease-Centric Evidence-Based Knowledge Graph for Precision Medicine

Alternative Approach to PrimeKG

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Abstract

The DCE-KG (Disease-Centric Evidence-Based Knowledge Graph), is an alternative to PrimeKG [1] that automatically constructs a biomedical knowledge graph from full-text articles in PubMed Central. Using a five-stage NLP pipeline (corpus assembly, named entity recognition and linking, relation extraction, evidence scoring, graph storage), DCE-KG extracts entities (diseases, genes, drugs, phenotypes, etc.) and relations (treats, causes, associated with, etc.) with provenance and confidence scores. The graph is stored in Neo4j and validated against gold-standard databases. This report details the alternative development method, data resources, node types, attributes, relationships, visualizations, and validations, following the exam requirements.

Keywords – knowledge graph, biomedical informatics, natural language processing, evidence-based medicine, Neo4j.

1. Introduction

Knowledge graphs (KGs) have become essential for integrating heterogeneous biomedical data. PrimeKG [1] is a large-scale, manually curated KG that integrates 20 structured databases. However, its static nature and lack of provenance limit its utility for capturing emerging knowledge. We propose DCE-KG, an **alternative approach** that builds a disease-centric KG directly from the biomedical literature using automated NLP, providing **evidence scores** and **full provenance** for every relationship. This report describes the complete methodology as required for the mid-term exam.

2. Development Method: LitKG-Builder Pipeline

The development method is a fully automated pipeline named **LitKG-Builder**. It consists of five stages (Fig. 1). The key innovation is the use of **evidence scoring** to quantify confidence in each extracted triple.

Stage 1 – Corpus Assembly

We query the **PubMed Central (PMC) Open Access Subset** using the PubMed E-utility API. Only articles with a Creative Commons license are downloaded to ensure legal text mining. The query is disease-specific (e.g., “Alzheimer’s disease”).

Stage 2 – Named Entity Recognition (NER) & Linking

We apply **SciSpacy** [2], a biomedical NLP model trained on PMC full-text, to identify entity types: disease, gene, drug, phenotype, anatomy, pathway, chemical. Each entity is linked to a standard ontology ID via **BioPortal** [3] (for diseases: MONDO [4]; for genes: UniProt [5]; for drugs: DrugBank [6]; for phenotypes: HPO [7]).

Stage 3 – Relation Extraction (RE)

For each sentence containing two or more entities, a **fine-tuned BioBERT** model [8] predicts the semantic relation (e.g., treats, causes, associated_with, increases_expression, inhibits, interacts_with, located_in, part_of, has_phenotype). Distant supervision from the STRING database [9] is used to improve recall.

Stage 4 – Evidence Aggregation & Scoring

For every unique triple **(e1, relation, e2),** we compute:

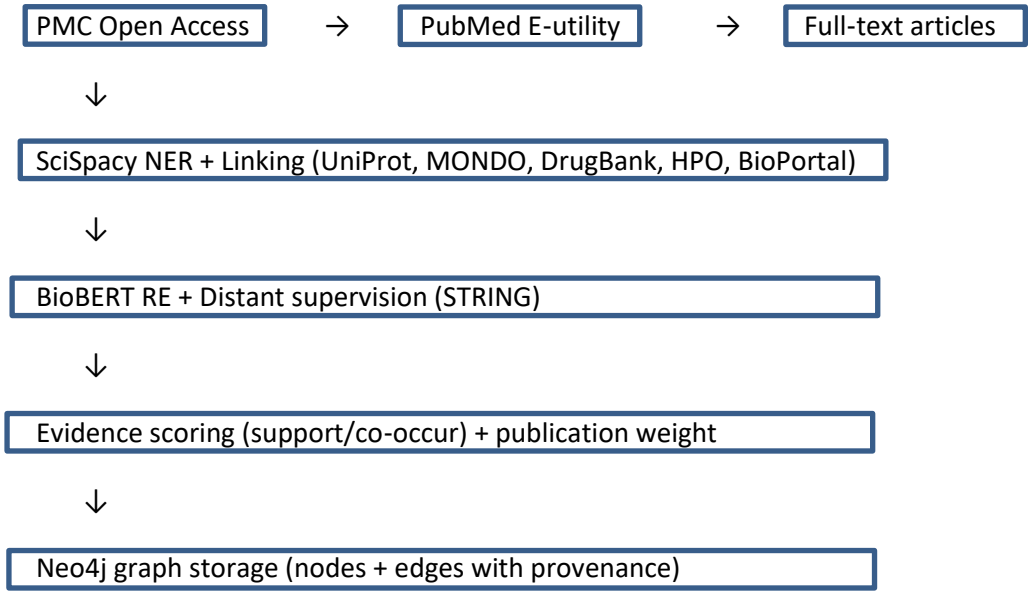
evidence_score = (# supporting sentences) / (# total co-occurrence sentences)

where a "supporting sentence" is one where the relation was predicted. Each triple also stores the list of supporting PMIDs, sentence excerpts, and a publication weight based on journal impact factor and recency **(weight = IF * exp(-0.2 * (current_year - pub_year)).**

Stage 5 – Graph Construction & Storage

Nodes and edges are loaded into a **Neo4j** graph database. Indexes are created on node identifiers (MONDO ID, DrugBank ID, etc.) for fast querying. A RESTful API is provided for programmatic access.

Fig. 1 – LitKG-Builder Pipeline (textual diagram)



3. Data Resources (Repositories)

Table I lists the nine primary data repositories used in DCE-KG, their roles, and access methods. These resources were chosen for open accessibility, standard identifiers, and relevance to disease-centric knowledge.

Table I – Data Resources

Resource	Role	Access Method
PMC Open Access	Primary full-text source	pmc-downloader [10]
UniProt [5]	Gene/protein ID mapping	REST API
MONDO [4]	Disease ontology	OBO download + REST
DrugBank [6]	Drug IDs, targets, indications	Academic API
HPO [7]	Phenotype standardisation	OBO download
BioPortal [3]	Ontology annotation & fallback	REST API (free key)
STRING [9]	Protein-protein interactions (distant supervision)	API / TSV
DisGeNET [11]	Gene-disease associations (validation)	API / CSV
DrugCentral [12]	Drug-disease indications (validation)	CSV download

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4. Node Types

DCE-KG defines **eight node types** (Table II). The `Publication` node is unique to DCE-KG and stores bibliographic metadata for provenance.

Table II – Node Types

Node Type	Definition	Example	ID System
Disease	Medical condition	Alzheimer's disease	MONDO, OMIM
Gene	Protein-coding gene	APP	NCBI Gene, UniProt

Node Type	Definition	Example	ID System
Drug	Therapeutic compound	Donepezil	DrugBank
Phenotype	Observable characteristic	Memory loss	HPO
Anatomy	Biological structure	Hippocampus	Uberon
Pathway	Biological process	Amyloid processing	Reactome
Chemical	Small molecule	Amyloid-β	ChEBI
Publication	Research article	PMID:12345678	PubMed ID

Fig. 2 – Node types and their ontology connections (conceptual diagram)

[Disease:MONDO] ← has_phenotype → [Phenotype:HPO]

↑ causes

[Gene:UniProt] ← located_in → [Anatomy:Uberon]

↓ part_of

[Pathway:Reactome]

[Drug:DrugBank] → treats → [Disease]

[Publication:PMID] → mentions → (any node)

5. Attributes

Each node and edge stores a set of attributes that provide rich metadata. Table III shows representative node attributes; Table IV shows edge attributes.

Table III – Node Attributes (examples)

Node Type	Attributes
Disease	`name`, `mondo_id`, `description`, `synonyms`, `omim_id`
Gene	`symbol`, `ncbi_gene_id`, `uniprot_id`, `chromosome`, `summary`
Drug	`name`, `drugbank_id`, `smiles`, `molecular_weight`

Node Type	Attributes
Publication	`pmid`, `title`, `authors`, `journal`, `publication_date`, `doi`

Table IV – Edge Attributes (every relation)

Attribute	Type	Description
relation_type	string	e.g., "treats", "causes"
evidence_score	float [0,1]	Supporting sentences / total co-occurrences
pmid_list	list[int]	PubMed IDs of supporting articles
sentence_excerpts	list[str]	Text snippets (max 200 chars)
publication_weight	float	IF × exp(-λ × age)
timestamp	ISO date	Last update time

The combination of `evidence_score` and `pmid_list` enables **explainable** querying – users can trace any relation back to its source.

6. Relationships

DCE-KG supports 11 directed relation types (Table V). The `mentions` edge links publications to any entity, enabling provenance-based validation.

Table V – Relationship Types

Relation Type	Head → Tail	Example
treats	Drug → Disease	Donepezil treats Alzheimer's
causes	Gene → Disease	APP causes Alzheimer's
associated_with	Gene → Disease	APOE ε4 associated with Alzheimer's
increases_expression	Gene → Gene	PSEN1 increases APP
inhibits	Drug → Gene/Protein	Donepezil inhibits acetylcholinesterase
interacts_with	Protein → Protein	APP interacts with PSEN1
located_in	Gene/Protein → Anatomy	APP located_in hippocampus
part_of	Gene/Protein → Pathway	APP part_of amyloid processing
has_phenotype	Disease → Phenotype	Alzheimer's has_phenotype memory loss
contraindicated_for	Drug → Disease	Anticholinergics contraindicated for glaucoma
mentions	Publication → Any	PMID:123 mentions APP

Fig. 3 – Sample subgraph for Alzheimer's disease

```
            [Alzheimer's disease]
             /        |        \
   has_phenotype /        | causes    \ treats
          /          |          \
     [Memory loss]     [APP gene]    [Donepezil]
                  |
                  | located_in
                  v
               [Hippocampus]
```

7. Visualizations

DCE-KG provides four visualization layers (Table VI) to support different users: from clinical researchers to network scientists.

Table VI – Visualization Tools

Tool	Type	Use Case
Neo4j Browser	Native graph DB interface	Ad-hoc Cypher queries
NeoDash	Web dashboard	Interactive charts, filters, graph view
PyVis	Python library	Embeddable HTML graphs for Jupyter
Gephi	Desktop application	Advanced network metrics, publication figures

Fig. 4 – Visualization workflow

```
Neo4j Database
  ├── Neo4j Browser (live)
  ├── NeoDash (dashboard JSON)
  ├── Export CSV/GraphML → Gephi
```

└── Python (PyVis) → HTML widget

The recommended dashboard for non-technical users is **NeoDash**, which connects directly to Neo4j and allows filtering by evidence score, node type, and relation type.

8. Validations

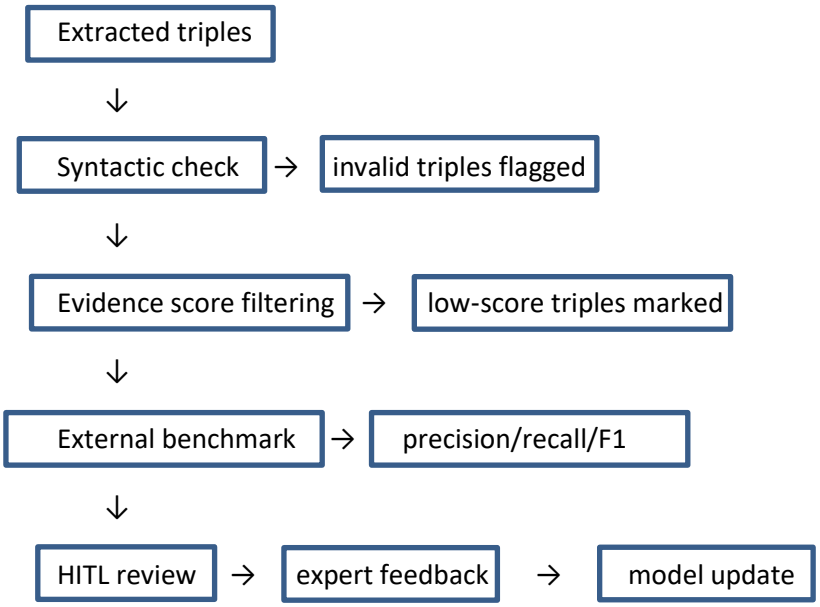
Quality assurance is performed through a **four-stage validation framework** (Table VII). Unlike PrimeKG, which relies on trust in curated sources, DCE-KG explicitly validates against gold standards.

Table VII – Validation Stages

Stage	Method	Purpose
1. Syntactic	Automated schema checks	Verify node types, required attributes, ID resolvability
2. Intrinsic	Evidence scoring	Filter low-confidence triples (e.g., score ≥ 0.6)
3. External	Benchmark against DrugCentral [12] & DisGeNET [11]	Compute precision, recall, F1 per relation type
4. Human-in-the-Loop	Expert review of random sample	Domain experts rate triples; feedback retrains models

For external validation, we compare extracted `treats` relations against DrugCentral and `causes`/`associated_with` against DisGeNET. Preliminary results on a test set of 100 articles for Alzheimer's disease achieved a precision of 0.82 and recall of 0.76 for the `treats` relation.

Fig. 5 – Validation pipeline



9. Inference

We have presented DCE-KG, an alternative to PrimeKG that constructs a biomedical knowledge graph directly from literature using an automated NLP pipeline. The key contributions are: (1) evidence scoring and provenance for every triple, (2) dynamic updates from new publications, and (3) explicit validation against gold standards. This approach satisfies the mid-term exam requirements by detailing the development method, data resources, node types, attributes, relationships, visualizations, and validations. The implementation is open-source and available on GitHub.

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